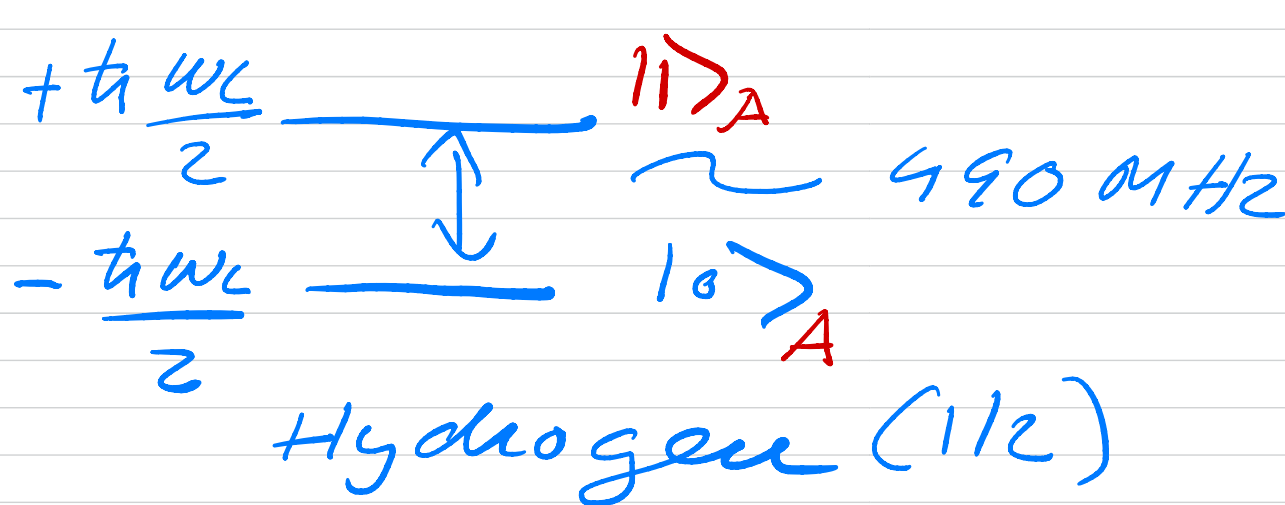


FYS5419/9419,
lecture February 12,
2025.

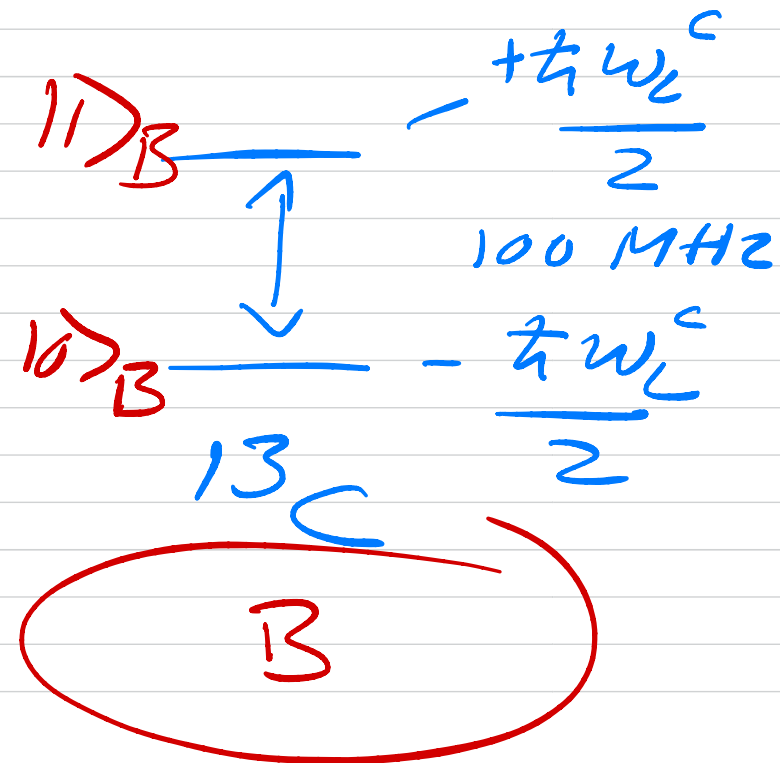
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$$|4(t)\rangle = U(t) |4(0)\rangle$$

$$U(t) = e^{i\hbar\omega_L t \sigma_z/2}$$



A $|0\rangle_A$



B

$$|E[S_x] = \langle S_x \rangle$$

$$= \frac{\hbar}{2} \left(\langle 0 | e^{-i\omega_c t/2} \cos \theta/2 + \right. \\ \left. \langle 1 | e^{i\omega_c t/2} e^{i\varphi_{nm} \theta/2} \right)$$

$$\times \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{\sigma_x} \left(e^{i\omega_c t/2} \cos \theta/2 |0\rangle + e^{-i\omega_c t/2} \sin \theta/2 e^{i\varphi_{nm}} |1\rangle \right)$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\sigma_x \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \sigma_x |0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ = |1\rangle$$

$$= \frac{\hbar}{2} \sin \theta \cos(\phi - \omega_L t)$$

$$\langle S_y \rangle = \frac{\hbar}{2} \sin \theta \sin(\phi - \omega_L t)$$

$$\langle S_z \rangle = \frac{\hbar}{2} \cos \theta$$

y-component oscillates as
the x-component but with a
 $\pi/2$ phase difference

Rotating wave approximation

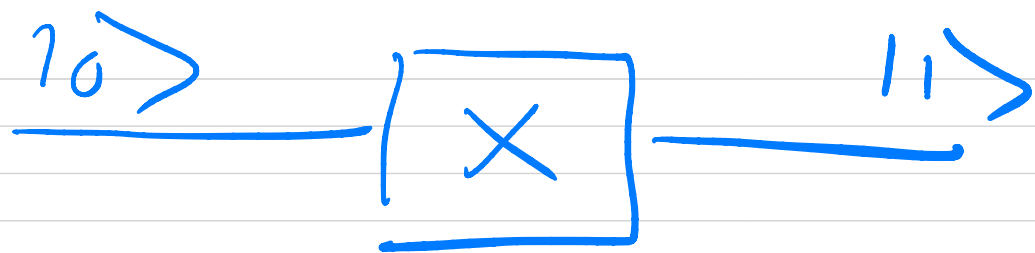
$$\vec{B}_{\pm}(t) = \frac{B_0 f}{2} (\cos(\omega_0 t) \vec{x} \pm \sin(\omega_0 t) \vec{y})$$

$$|\psi'(t)\rangle = \hat{R}_Z(\omega_0 t) |\psi(t)\rangle$$

$$\downarrow$$
$$e^{-i\omega_0 t \nabla_z / 2}$$

$$H'(t) \approx -\frac{\hbar \omega_R}{2} \nabla_x = -\frac{\hbar \omega_R}{2} X$$

$$\nabla_x = X \quad \nabla_y = Y \quad \nabla_z = Z$$



(NOT gate)

$$\hat{R}_x = e^{i\omega t \nabla_x / 2}$$

$$\hat{R}_x |0\rangle = |1\rangle$$

$$\hat{R}_x(\pi/2) \hat{R}_y(\theta) \hat{R}_x(-\pi/2)$$

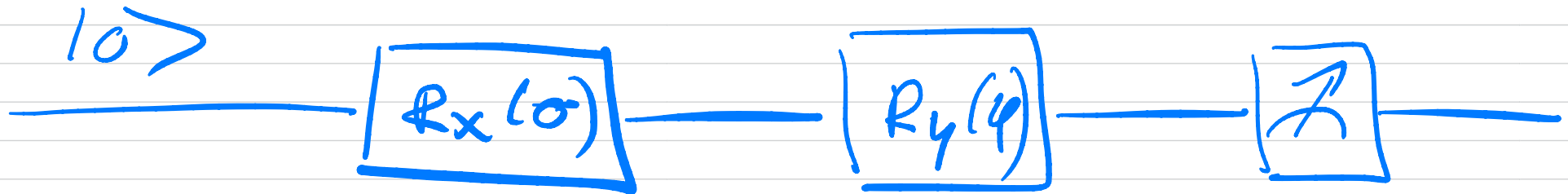
$$= \frac{1}{\sqrt{2}} (1 - i\nabla_x) \left(\cos \frac{\theta}{2} 1 - i \sin \left(\frac{\theta}{2} \right) \nabla_y \right) \times \frac{1}{\sqrt{2}} [1 + i\nabla_x] = \cos \left(\frac{\theta}{2} \right) 1 - i \sin \left(\frac{\theta}{2} \right) \nabla_z$$

$$= R_z(\theta)$$

comp basis

$|0\rangle$ and $|1\rangle$

$$|4\rangle = \alpha |0\rangle + \beta |1\rangle$$



$$\hat{H} = cI + dZ + eX$$

$\uparrow$
 HZH